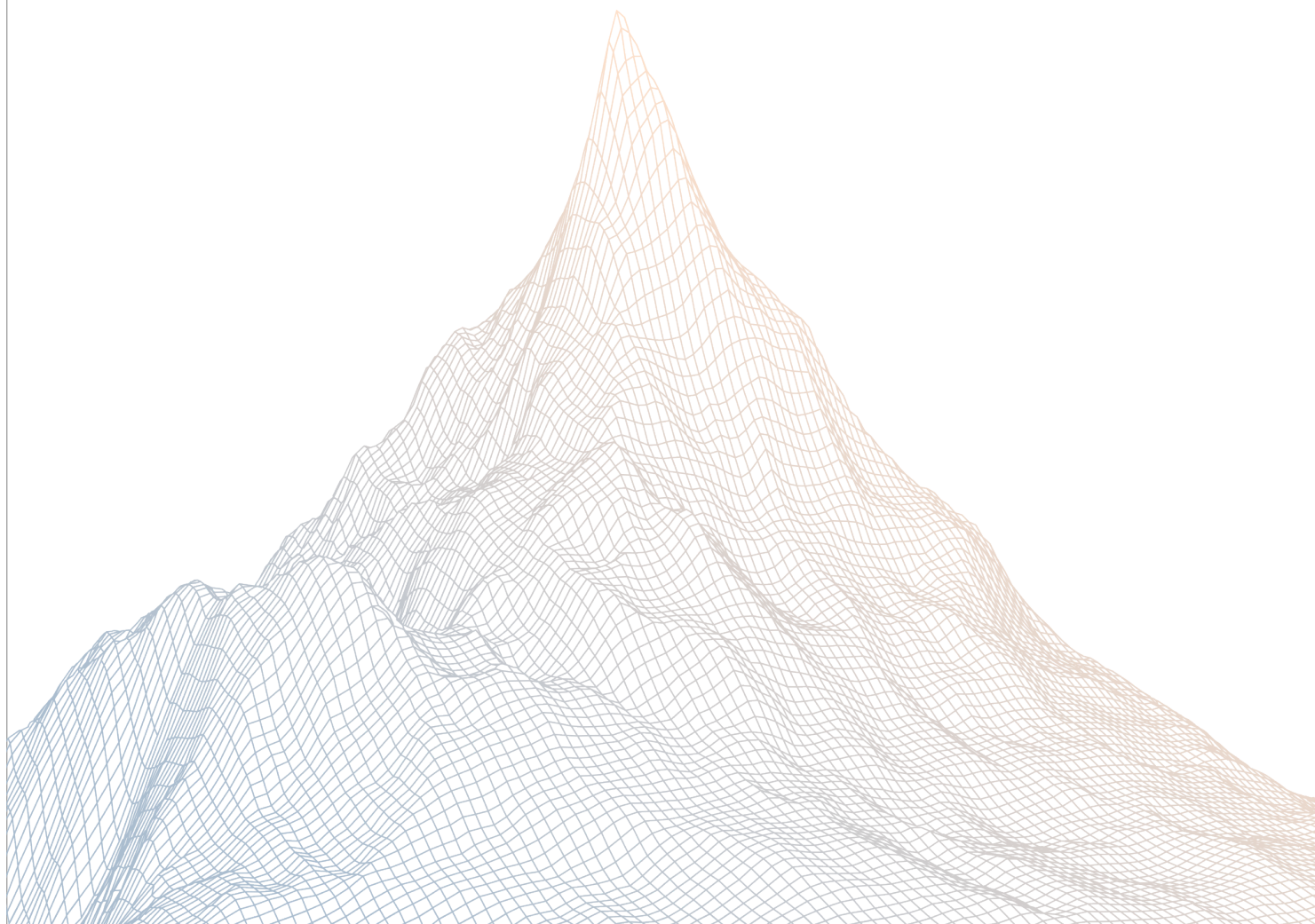


Meteora

Smart Contract Security Assessment

VERSION 1.1



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Introduction

1.1 About Zenith

Zenith assembles auditors with proven track records: finding critical vulnerabilities in public audit competitions.

Our audits are carried out by a curated team of the industry's top-performing security researchers, selected for your specific codebase, security needs, and budget.

Learn more about us at <https://zenith.security>.

1.2 Disclaimer

This report reflects an analysis conducted within a defined scope and time frame, based on provided materials and documentation. It does not encompass all possible vulnerabilities and should not be considered exhaustive.

The review and accompanying report are presented on an "as-is" and "as-available" basis, without any express or implied warranties.

Furthermore, this report neither endorses any specific project or team nor assures the complete security of the project.

1.3 Risk Classification

SEVERITY LEVEL	IMPACT: HIGH	IMPACT: MEDIUM	IMPACT: LOW
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

2

Executive Summary

2.1 About Meteora

Our mission is to build the most secure, sustainable and composable liquidity layer for all of Solana and DeFi.

By using Meteora's DLMM and Dynamic AMM Pools, liquidity providers can earn the best fees and yield on their capital.

This would help transform Solana into the ultimate trading hub for mainstream users in crypto by driving sustainable, long-term liquidity to the platform. Join us at Meteora to shape Solana's future as the go-to destination for all crypto participants.

2.2 Scope

The engagement involved a review of the following targets:

Target	damm-v2
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Repository	https://github.com/MeteoraAg/damm-v2
-------------------	---

Commit Hash	9d2fcd5353935fdee44c244b23fbcf7dc1d8c432
--------------------	--

Files	Changes in PR-94
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2.3 Audit Timeline

September 10, 2025	Audit start
September 12, 2025	Audit end
October 15, 2025	Report published

2.4 Issues Found

SEVERITY	COUNT
Critical Risk	0
High Risk	0
Medium Risk	0
Low Risk	1
Informational	0
Total Issues	1

3

Findings Summary

ID	Description	Status
L-1	Partial fill allows for 0 value swap	Resolved

4

Findings

4.1 Low Risk

A total of 1 low risk findings were identified.

[L-1] Partial fill allows for 0 value swap

SEVERITY: Low

IMPACT: Low

STATUS: Resolved

LIKELIHOOD: Low

Target

- src/instructions/swap/swap_partial_fill.rs#L27-L40

Description:

When the partial fill swap is used the actual amount in is reduced if the swap would go above the max or below the min.

```
let max_amount_in = get_delta_amount_a_unsigned(  
    self.sqrt_min_price,  
    self.sqrt_price,  
    self.liquidity,  
    Rounding::Up,  
)?;  
  
let (consumed_in_amount, next_sqrt_price) = if amount_in ≥ max_amount_in {  
    (max_amount_in, self.sqrt_min_price)  
}
```

However if we are already at the max/min, this will allow us to swap a 0 amount which is not intended and checked for on all other swap paths.

```
let excluded_transfer_fee_amount_in =  
    calculate_transfer_fee_excluded_amount(token_in_mint,  
    amount_in)?.amount;  
  
require!(excluded_transfer_fee_amount_in > 0, PoolError::AmountIsZero);  
  
let swap_result = pool.get_swap_result_from_partial_input(  

```

```
        excluded_transfer_fee_amount_in,  
        fee_mode,  
        trade_direction,  
        current_point,  
    )?;  
  
    // @audit - I think this could be 0  
    let excluded_transfer_fee_amount_out =  
        calculate_transfer_fee_excluded_amount(token_out_mint,  
        swap_result.output_amount)?.amount;  
  
    require!(  
        excluded_transfer_fee_amount_out ≥ minimum_amount_out,  
        PoolError::ExceededSlippage  
    );
```

Recommendations:

We recommend adding a check that ensures that
`swap_result.included_fee_input_amount > 0`

Meteora: Resolved with [PR-113](#)

Zenith: Verified